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MILITARY SPECIFICATION

MANUALS, TECHNICAL: AIRBORNE MISSILES AND GUIDED WEAPONS, PREPARATION OF

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification prescribes the technical content requirements for the preparation of technical manuals for intermediate (Fleet), Naval Weapons Station (NAVWPNSTA) and depot maintenance of airborne missiles and guided weapons.

1.2 Types of Manuals. Technical manuals prepared to this specification shall be of the following types: Intermediate (Fleet) Maintenance with Illustrated Parts Breakdown, NAVWPNSTA Maintenance with Illustrated Parts Breakdown or Depot Maintenance with Illustrated Parts Breakdown.

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications, Standards, and Publications. The following specifications, standards and publications form a part of this specification to the extent specified herein. Unless otherwise specified in the contract, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and the supplement there to, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-81927	Manuals, Technical: General Style and Format of (Work Package Concept)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standards Department (SESD), ATTN: Code 53, Lakehurst, NJ 08733-5100 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-M-81929	Manuals, Technical: Illustrated Parts Breakdown; (Work Package Concept) Preparation of
MIL-M-85337	Manuals, Technical: Quality Assurance Program; Requirements For

PUBLICATIONS

NAVY

OPNAVINST 4790.2	Naval Aviation Maintenance Program
OPNAVINST 8600.2	Naval Airborne Weapons Maintenance Program
NAVAIR 00-25-700	Technical Manual Preparation Guide for Technical Writers, Editors, and Illustrators
NAVAIR 11-1-116/ TW010-AA-ORD-030	Navy Ammunition Logistics Codes

(Copies of specifications, standards, drawings and publications required by preparing activities in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

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3.1 Style and Format. The style and format of manuals prepared to this specification shall be in accordance with MIL-M-81927.

3.2 General. Except when noted for specific application or restricted from use in a specific application, arrangement of manuals shall be in accordance with MIL-M-81927. Technical manuals prepared in accordance with this specification shall follow the Work Package (WP) concept and technical manual planning and organizational techniques described in NAVAIR 00-25-700. The WP's shall contain general and specific instructions for intermediate and depot levels of maintenance and, when applicable, special instructions for missile or guided weapon processing in NAVWPNSTA and NADEP. Whenever possible, specific maintenance instructions shall be standardized to ensure uniform, systematic processing methods at all maintenance activities while maintaining highest safety standards. Nomenclature, Cautions and Notes for similar equipment shall be consistent throughout the manual.

3.3 Technical Manual Management. A separate manual shall be prepared for each level of maintenance. When directed by the requiring activity, a combined NAVWPNSTA/Depot Maintenance manual may be prepared (see 6.2). If the manual covers both NAVWPNSTA and Depot maintenance, the maintenance level identification shall appear in the WP title block and in the text. For example, if the complete task is restricted to performance by NAVWPNSTA personnel, it shall be contained in a separate WP and the words NAVWPNSTA MAINTENANCE shall appear in the WP title block (see figure 1). If a portion of the task is restricted to a specific maintenance level, those procedures shall be contained in separate paragraphs. The paragraph title shall identify the restriction, for example REMOVAL OF THERMAL BATTERY (NAVWPNSTA). If similiar restrictions exist between Fleet - shore - based and Fleet - ship board maintenance, they shall be identified as noted above. An explanation of such locations shall be continued in the introduction WP.

3.3.1 Industrial Engineering Techniques. Sound engineering principles, performance data, and available reliability data shall be used in the preparation of maintenance procedures. WP's prepared for use by intermediate and depot personnel shall include industrial engineering techniques that place emphasis on performance of tasks.

3.3.2 Standard Shop Practices. Standard shop practices and techniques detailed in general engineering series manuals shall not be duplicated. When required, appropriate reference shall be made to the applicable general series publication.

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3.4 Inspection Procedures. (See figure 11) In addition to the requirements of MIL-M-81927, certain procedures in Intermediate (Fleet) Maintenance manuals only will require inspection by the user. Inspection shall be required for all procedures that, if improperly performed, could reduce the usability of the unit or product, cause weapon or equipment failure, or jeopardize the safety of personnel. Inspection procedures shall define the characteristics to be inspected; procedures, equipment, gages, etc. to be used; minimum inspection requirements; and sound acceptance and rejection criteria. Inspections shall be described in sufficient detail to ensure that the product conforms to technical requirements. Certain inspections contained in the Fleet manual are accomplished by designating inspectors. These inspections shall be identified by the abbreviation (QA) following the step as follows:

m. Install new screws and torque to 13 ± 1 inch-pounds. (QA)"

3.5 Procedural Steps. Every action that furthers the accomplishment of the production effort shall be expressed as a step. Procedural steps shall begin with an action verb and direct accomplishment in a logical, safe, simple and direct manner. Procedural steps shall be in accordance with MIL-M-81927.

3.5.1 Procedural Steps - Production Actions. It shall be assumed that each procedure will be performed by the lowest skill level consistent with good industrial engineering practices and safety. The detailed step shall include such amplifications as required to ensure that personnel performing the task need not refer to any supplementary instructions except where such reference material is lengthy.

3.6 Technical Content - General. Except where noted for specific application or restricted from use in a specific application, the following guidelines apply to all manuals prepared to this specification.

3.6.1 Support Equipment Required Lists. (See Figure 2.) Support equipment required to perform operational checks or maintenance shall be listed prior to the procedures to which the SE applies. Only those tools and equipment authorized for use at the level of maintenance covered shall be listed. If the WP contains multilevel maintenance procedures, and one or more of the SE items listed is authorized for use only at one level, this restriction shall be indicated by the use of an asterisk (*) following the item nomenclature. An appropriate notation shall follow the listing. For example: "* Authorized for use at depot level only." Standard hand tools shall not be listed. However, standard measuring equipment, such as voltmeters and torque wrenches shall be listed. All special tools and test equipment shall be listed in alphabetical sequence by noun nomenclature, and type designation or part number and manufacturer's code. Items shall be listed in column format under the heading "Support Equipment Required". The quantity of items required, if applicable, shall follow

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the nomenclature in parentheses. Illustrations shall not be prepared in support of such lists, but may be developed in support of maintenance procedures. For example: as required to illustrate the fabrication of inter-connecting cables or other local manufacture items of this category.

3.6.2 Materials Required List. (See Figure 2.) Materials required lists shall be included when consumable materials and expendable items are required to perform work in WP procedures. Materials required list shall follow immediately after the support equipment required list (if applicable) in WP's containing the procedures in which the material is used. Items shall be listed in alphabetical sequence, by nomenclature and specification or part number. The quantity of items required shall follow the nomenclature in parentheses. Material will always be listed by Government specification unless alternate identification is approved by the requiring activity.

3.6.3 Emergency Procedures. Information and procedures relative to emergencies shall be provided. When applicable, this information shall be provided by the Government for inclusion in the manual. The following information shall be provided; however, these items shall be interpreted as minimum coverage:

a. Rendering-safe and disposal emergency procedures that can be performed by ordnance personnel, prior to the arrival of explosive-ordnance personnel, to alleviate or eliminate a hazardous condition that, if unabated, could lead to destruction of property and serious injury or loss of life.

b. Peculiar firefighting techniques.

c. Tabulation of cookoff times and personnel evacuation distances under firefighting conditions (see figure 3).

d. Procedures required under accident or incident conditions involving dropping, collision, or inadvertent release.

e. Procedures to be followed when the missile or guided weapon has been exposed to adverse weather conditions.

3.7 Arrangement of Technical Content Work Packages. Generally, the arrangement of the technical content (WP 003 00 and subsequent) shall follow the required breakdown of the end item, its sections, and components. That is, those WP's associated with the assembled missile or guided weapon shall be presented first; for example, description, principles of operation, preparation for use or test, testing and troubleshooting, maintenance with IPB, preparation for shipment or storage, etc. of the missile or guided weapon. The sections of the end item would be similarly covered, except principles of operation would not be required, and each section would be followed by its component parts before the next section

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would be covered. The number of WP's required for coverage of each functional task and supporting data will vary dependent on the design and complexity of the missile or guided weapon. In addition, the maintenance concept will affect the number of WP's required.

3.7.1 Description. A brief physical description of the missile or guided weapon, the function of its basic elements, and the purpose of the weapon shall be provided. Technical characteristics and other general information required for the performance of maintenance shall be provided. If the manual covers more than one model, the significant differences among the models shall be explained. Detailed physical descriptions of components shall not be required. A table of characteristics shall list the weight, center of gravity, power requirements, explosive weight, and class of the missile or guided weapon.

3.7.2 Principles of Operation. A functional description of the end item and its sections and functional components shall be provided. These principles of operation shall be provided only to the depth required to support the maintenance concept. The principles of operation shall serve as a guide for the technician as to section and component nomenclature, and the general arrangement of the end item. Nomenclature shall remain consistent throughout this WP(s) and all other WP's contained in the manual. The intent of the principles of operation is to support the qualified technician with reference material that may be required in the performance of the task and to support on-the-job training of less-qualified personnel.

3.7.2.1 Principles of Operation - Method of Presentation. A breakdown of the end item by sections shall be presented first. All configurations of the end item shall be presented, including components such as motors, warheads, fuzes, and associated accessories. Configuration data shall be presented in a table (figure 4). Identification markings shall be shown exactly as they will be found on the missile or guided weapon. The functional description shall include the purpose, type, content, and main features of each section. Major characteristics, capabilities, and limitations shall be provided. Characteristics which are classified information shall not be included unless absolutely necessary for clarity of presentation. If required, these characteristics should be contained in a classified supplement to the manual. Refer to MIL-M-81927. Locator illustrations of components as installed in the end item shall be provided. See figure 5. Bit-and-piece parts shall be illustrated only when required to support the functional description. Internal parts of components shall not be described unless specifically required to understand system function. For example, internal part description of a pump would not be required when indication of a pressure supply is all that is required to describe that part of the system operation.

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3.7.3 Preparation for Use, Test, or Issue. Procedures necessary to prepare the missile or guided weapon for use, test, or issue, including unpacking, inspection, and assembly, shall be provided as applicable. Criteria for accepting or rejecting components, authorized adjustments or replacement of components, and instructions for the disposing of defective items shall be included.

3.7.3.1 Unpacking. Instructions for unpacking shall include the following as applicable: checking environmental indicators; depressurizing shipping containers; depalletizing, uncrating, or decanning the end item or its components; depreserving (for example, removing protective covers and coatings peculiar to the end item or its components) and inspecting containers.

3.7.3.2 Inspection. Procedures for visual, electrical, electronic, and mechanical inspections, and test of components to certify them ready for assembly shall be provided in tabular format. Reference shall be made to testing and troubleshooting WP's when applicable. Specific conditions, such as indications of rough handling, corrosion, loose or punctured weather seal on motor nozzle, and damaged wiring or threads, shall be included. Classification of characteristics established to provide for detection of defects (dents, gouges, cracks, burrs, etc.) shall be designated as either CRITICAL, MAJOR, or MINOR (See 6.3). The inspection tables in NAVWPNSTA manuals only shall include a column for defect classification (See figure 6). The preparing activity may recommend a classification of characteristics; however, these recommendations shall be approved by the cognizant engineering activity. Procedural steps shall not contain classification characteristics. When applicable, values, tolerances, and support equipment requirements shall be provided. Inspection steps shall include the maintenance action required, either in text or by reference, if a defect is found. For example:

d. Inspect guidance section in accordance with table 1. Section markings are illustrated in figure (4). Inspection points are illustrated in figure 5 (See figure 6).

or

e. Inspect heat exchanger for corrosion. If corrosion is found, remove and replace heat exchanger (WP 069 00).

3.7.3.3 Assembly. Instructions for assembly of components of each section of the end item and assembly of the sections to build up the complete missile or guided weapon shall be provided. The scope of assembly instructions shall reflect the configuration of components and sections of the end item as normally delivered to the level of maintenance covered. All configurations shall be covered. Assembly instructions shall include all pertinent assembly criteria,

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including clearances, torque values, and similar data. Instructions for adjustment and alignment shall be provided in logical sequence. Instructions for lubrication, application of corrosion-preventive compounds, gasket pastes, etc., required during assembly, shall be provided. Testing of the assembled missile or guided weapon shall be referenced to the applicable testing and troubleshooting WP.

3.7.4 Testing and Troubleshooting. (See Figure 7.) Testing and troubleshooting procedures shall be developed to the extent required to maintain the equipment at the authorized maintenance level. Troubleshooting data shall be presented in combination with test procedures. These combinations shall present the procedures required to guide the technician, in as practical a manner as possible, to the cause of failure or malfunction and its repair. All information required to perform the test and evaluate probable malfunctions of the assembled missile or guided weapon shall be provided. When applicable, this information shall include all modes of operation. Testing and troubleshooting data shall be provided for the sections of the end item and their repairable components. Refer to 3.7. The number of WP's required will depend on production planning and control, and step-by-step sequence requirements for accomplishment of each maintenance action at the appropriate maintenance level from disassembly or receipt of individual sections or components to complete assembly of components, sections, and the end item.

3.7.4.1 Testing and Troubleshooting Procedures. (See Figure 7.) Testing and troubleshooting procedures shall include determination of the cause, and isolation of the fault to replaceable parts, interface wires, or mechanical linkage. The procedures shall contain the following:

a. Identification of the missile or guided weapon, section, or component to be covered.

b. A concise explanation of the troubleshooting peculiar to the WP.

c. A simplified locator illustration of support equipment and an associated legend or table which shall locate each component, accessory, connector, or junction box in the item under test and each connector used as a test point.

d. General maintenance procedures and precautions applicable to testing and troubleshooting.

e. Controls and displays illustrations, as required.

f. Pretest setup procedures, including setup and use of support equipment during the test (see figure 8).

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g. Power turn-on procedures.

h. Test procedures, including instructions required for use of support equipment.

3.7.4.1.1 Test procedures shall take into account accessibility of test points, controls, indicators, and the feasibility of using built-in test capabilities, when available. The documentation of procedures shall be based on the following assumptions.

a. Only one malfunction exists at a time.

b. Successful completion of the test verifies system operation.

c. If a test result is not observed as called out in the test procedure, the malfunction symptom shall be keyed to the appropriate part of the trouble-shooting diagram. However, if there is only one remedy for a specific malfunction, the remedy may appear following the appropriate test procedural step.

d. When a component is replaced, the test should be restarted to ensure that the replacement did not introduce a new fault into the system.

e. If several components are suspect, the one most likely to have failed shall be considered first; however, consideration shall be given to access and ease of replacement when the choice of the component is not firm.

3.7.4.2 Testing and Troubleshooting Data. These instructions shall include what data is to be recorded, what forms or other recording vehicle is to be used, and what reporting is required.

3.7.5 Maintenance. Procedures shall be provided for the maintenance and rework of the missile or guided weapon and its sections and components at the applicable level of maintenance. These procedures shall include the subjects covered in the following paragraphs as they apply to the end item, section, or repairable component, and as required by the maintenance plan.

3.7.5.1 Related Tasks. WP's prepared for maintenance or rework shall include related tasks. Related tasks are any production actions related to the rework or overhaul of the missile or guided weapon which will yield a savings in resources when scheduled for accomplishment in conjunction with the WP concerned. It is assumed that these tasks will appear in two or more WP's. Dependent on the length of the procedure in questions, they shall be included either by duplication or reference. Refer to MIL-M-81927 for guidelines. Related tasks shall be included when there is a definite place in the overall task when the

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procedure can best be accomplished, or when it must be accomplished to complete the task. The purpose of identifying a related task is to cause several tasks to be accomplished at the same time, thereby eliminating unnecessary assembly or disassembly, and reducing the possibility of the entrance of foreign material. Because of the multiple maintenance actions required during rework and overhaul, and the various possible configurations of the end item, it is possible that there may be several instances when a related task will be "best accomplished". The Task Identification Matrix and the associated task analysis of the rework or overhaul cycle must be performed in accordance with NAVAIR 00-25-700 to ensure related tasks are properly scheduled and not inadvertently repeated in any one cycle.

3.7.5.2 Disassembly. Procedures required for disassembly of the end item, its sections, and repairable components to the extent authorized at the maintenance level covered shall be provided. When possible, illustrations prepared for the illustrated parts breakdown (IPB) shall also be used for maintenance. Detailed illustrations required and not available in IPB figures shall be prepared. IPB illustrations may be reindexed in accordance with MIL-M-81927.

3.7.5.3 Cleaning, Inspection, and Repair. Instructions for cleaning that are peculiar to the end item, its sections, and components, specifying use of approved materials, shall be provided. Precautions to be observed shall be included. Inspection methods, equipment required, and instructions for complete inspection of component parts shall be provided. Allowable service limits and adequate standards for determining when parts should be repaired shall be provided. Instructions for repair shall be provided to the extent authorized at the maintenance level covered. Tables of acceptable service tolerance limits shall be provided. The tables shall serve as a standard and give the plus or minus tolerances that are acceptable for continued operation, and will not affect the reliability of the component or equipment. Normally, cleaning, inspection, and repair procedures shall be included in maintenance WP's.

3.7.5.4 Assembly. Procedures required for assembly of the end item, its sections, and repairable components that have been disassembled in accordance with the instructions provided in 3.7.5.2, shall be provided. The requirements for disassembly as stated in 3.7.5.2 also apply to reassembly.

3.7.6 Handling, Packing for Shipment, and Stowage. Procedures for handling, packing for shipment, and stowage of the missile or guided weapon, its sections, and components shall be provided, as applicable.

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3.7.6.1 Handling. Handling procedures for the end item, its sections, and its components shall be provided. Procedures peculiar to shipboard and shore-based activities shall be covered separately. Step-by-step procedures shall be provided for each operation. Hazards likely to be encountered shall be highlighted by the use of warnings or cautions as applicable. Detailed instructions on the use of common support equipment shall not be provided; however, instructions shall be provided if particular configurations of support equipment are required. For example, if specific adapters are used with the support equipment to handle the end item, detailed instructions and figures shall be provided. Lift points, hard points, and center of gravity measurements shall be provided as applicable. Special handling procedures for transporting to and from ready-service stowage shall be included, emphasizing the use of handling equipment.

3.7.6.2 Stowage. Stowage requirements such as temperature, humidity, and grounding shall be provided as necessary for magazine, open-air, and ready-service stowage. The ready-service stowage procedure shall cover stowage in shipboard and shore-based ready-service magazines, including NAVWPNSTA, dockside, and field bases. Illustrations may show typical ready-service magazines.

3.7.6.3 Packing for Shipment. Packing for shipment procedures for the end item, its sections, and its components shall be provided. These instructions shall include disposition actions, palletizing, types of containers to be used, marking of containers, and hazards likely to be encountered.

3.7.7 Local Manufacture Items. Procedures and illustrations for fabrication of items source coded as make from or assembled from shall be included. Items whose manufacture or assembly is simple and are adequately depicted in the IPB shall not be included. Depot "make from" items shall not be included.

3.8 Illustrated Parts Breakdown. The IPB for the end item, its sections, and its components shall be prepared in accordance with MIL-M-81929. When prepared as part of the WP, the IPB shall follow the maintenance instructions to which it applies. When prepared as a separate WP, the IPB shall follow the maintenance WP to which it applies.

3.8.1 Illustrations. Illustrations shall be prepared in accordance with MIL-M-81927 and the specific technical content requirements defined herein.

3.8.2 Diagrams. Diagrams and associated data required during maintenance shall be provided and shall include information necessary for maintenance personnel to accomplish the following:

a. Understand the makeup and function of each power, control, and signal circuit interface and mechanical system.

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AUR), but this capability is not normally used at the diagnostic BIT shall be run to check connector

b. Trace each circuit and mechanical system.

c. Accomplish testing and troubleshooting on circuits and mechanical equipment.

3.8.2.1 Functional Flow Diagrams. (See Figure 9.) Functional flow diagrams shall be prepared in support of principles of operation and as technical backup material for testing and troubleshooting procedures. The functional flow diagram shall provide a closed-loop representation of the operation of the missile or guided weapon. Point-to-point inter-relationship from one section to another shall be shown.

3.8.2.2 Point-to-Point Wiring Diagrams and Wire Lists. Point-to-point wiring diagrams, wire lists, or a combination of wiring diagrams and wire lists, shall be provided. The diagrams and wire lists shall be prepared in accordance with MIL-M-81927. Both a wire list and a point-to-point wiring diagram shall not be provided in support of the end item or any one section or component of the missile or guided weapon. Wire lists shall be supported by wiring diagrams. Point-to-point wiring from one component to another shall be shown. Wire numbers may appear on the diagram when required to ensure clarity of presentation. Input-output signals between controls and indicators shall be included. Test points, noting the accuracy and resolution of all signals; wave shapes, including scope settings if applicable; controls and indicators referred to in troubleshooting procedures; and mechanical linkage shall be illustrated as required. The basic layout of the diagram shall present signal flow from left to right, starting at a power source of mechanical input, and terminating at a display, surface control, etc.

3.9 Intermediate (Fleet) Maintenance Manual with Illustrated Parts Breakdown. Intermediate maintenance procedures, parts data, and processing methods for missiles and guided weapons, their sections, and components shall be provided in accordance with 3.6 through 3.8.2.2. Only those procedures authorized at this maintenance level shall be documented. The requirements defined in this specification are applicable to the degree required by the complexity of the missile or guided weapon and its associated equipment, and are not intended to restrict additional coverage that may be required.

3.9.1 Introduction Work Package - Intermediate (Fleet) Maintenance Manual. The introduction WP shall be provided in accordance with MIL-M-81927 and the following specific requirements. The introduction shall identify the missile or guided weapon by official nomenclature, briefly summarize the purpose and use of the end item in general terms, and list the cognizant and participating field activities responsible for the maintenance and rework of the missile or guided weapon. When the manual contains Department of Defense Identification Codes or

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Navy Ammunition Logistics Codes reference shall be made to NAVAIR 11-1-116/TW010-AA-ORD-030. An explanation of the identification of tasks or portions thereof restricted to specific maintenance activities shall be included. Refer to 3.3. Maintenance actions requiring the log book entries shall be identified in accordance with OPNAVINST 8600.2 or OPNAVINST 4790.2 as applicable. A historical Record of Applicable Technical Directives, complete lists of support equipment required, materials required and reference material shall be provided. Information on Quality Assurance procedures shall be included. The maintenance concept established for the missile and guided weapon shall be briefly discussed.

3.9.2 Processing Flow Chart. When specified by the requiring activity, one or more missile or guided weapon processing flow charts shall be provided (see 6.2.1). The chart shall be keyed to WP numbers and the test station at which the task is to be performed (see figure 10).

3.10 Depot Maintenance Manual with Illustrated Parts Breakdown. Depot maintenance procedures and parts data for sections or components of the missile or guided weapon shall be provided in accordance with 3.6 through 3.8.2.2. Only those procedures authorized at this maintenance level shall be documented. Procedures provided shall be detailed to the extent necessary to complete coverage for depot maintenance. The data shall be compatible with documentation provided in the Intermediate Maintenance Manual prepared in accordance with 3.9. The depot-level procedures shall not duplicate those provided in the Intermediate Maintenance Manual, except as required for clarity.

3.10.1 Introduction Work Package - Depot Maintenance Manual. The introduction to the manual shall be provided in accordance with MIL-M-81927 and the following requirements. The introduction shall identify the missile or guided weapon that the section or component to be overhauled is associated with. Identification shall be by official nomenclature. Maintenance actions requiring log book entries shall be identified in accordance with OPNAVINST 4790.2 or OPNAVINST 8600.2 as applicable. A historical Record of Applicable Technical Directives, complete list of support equipment required, materials required and reference material shall be provided. Information on Quality Assurance procedures shall be included as applicable.

3.10.2 Reference to Intermediate Maintenance Manual Work Packages. Maintenance tasks required to complete the depot-level maintenance that are contained in an Intermediate Maintenance Manual (for example, test of the assembled article after overhaul) shall be referenced by publication number and WP number. Short procedures contained in the Intermediate Maintenance Manual may be duplicated when required for safety, clarity, or continuity of task performance. Refer to MIL-M-81927 for guidelines.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for Inspection. Unless otherwise specified in the contract, the preparing activity is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the preparing activity may use their own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements (see 6.2.1).

4.2 All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the preparing activities' overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the preparing activity of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.3 In-Process Review, Validation, and Verification. The in-process review, validation, and verification of technical manuals shall be as directed by the requiring procuring activity and in accordance with MIL-M-85337. Validation and verification of the technical manual shall be completed prior to the implementation of tasks required at The NAVWPNSTA or DEPOT.

5. PACKAGING

5.1 Packaging Requirements. The requirements for packaging shall be in accordance with MIL-M-81927.

6. NOTES

6.1 Intended Use. Technical manuals prepared in accordance with the technical content requirements of this specification are intended for use in the maintenance of airborne missiles, guided weapons, and components at the intermediate (Fleet), NAVWPNSTA, and NADEPS.

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6.2 Ordering Data.

6.2.1 Procurement Requirements. Procurement documents should specify:

a. Title, number and date of this specification.

b. Whether a combined maintenance level manual is required (See 3.3).

c. Whether a processing flow chart is required (see 3.9.2).

d. Responsibility for inspections (see 4.1).

6.3 Definitions.

6.3.1 Critical. A critical characteristic is one that analysis indicates is likely, if defective, to create or increase a hazard to human safety or to result in failure of a weapon system or major system to perform a required mission.

6.3.2 Major. A major characteristic is one that analysis indicates is not critical, but is likely, if defective, to result in failure of an end item to perform a required mission.

6.3.3 Minor. A minor characteristic is one that analysis indicates is significant to product quality, but is not likely, if defective, to impair the mission performance of the item.

6.4 Figures Contained in this Specification. The figures illustrated in this specification are examples intended to illustrate style, format and sample content. They shall not be used for interpretation of specific technical content or exact scale requirements.

6.5 Subject Term (Key Word) Listing.

Airborne Missile
Fleet Maintenance
Guided Weapon
Military Specifications
Naval Weapons Station
Requirements
Technical Manuals

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6.6 Changes From Previous Issue. Asterisks (or vertical lines) have not been used in this revision to identify changes with respect to the previous issue. This is due to the extensiveness of the change.

Preparing Activity:

Navy - AS

(Project TMSS N218)

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NAVAIR 01-AIM9L/M-2.1
1 January 1986

021 00
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NAVWPNSTA MAINTENANCE
TESTING
ROCKET MOTOR
MK 36 MODS 7/8/9/10/11

Reference Material

Description, Safety and Emergency Procedures, Guided Missile and Training-Firing Missile AIM-9L/M and NATM-9L/M WP003 00
Maintenance with Illustrated Parts Breakdown Rocket Motor Mk 36 Mods 7/8/9/10/11 . . . . . WP022 00
Radiographic Inspection, Rocket Motor Mk 36 Mods 7/8/9/10/11 . . . . . WP023 00
Test Set, AN/DSM-98 (V) Guided Missile Electrical Circuit, Service, Calibration, and Alignment Instructions . . . . . NAVAIR 16-30DSM98-1
Unpacking, Inspection, and Packaging, Rocket Motor Mk 36 Mods 7/8/9/10/11 . . . . . WP020 00

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Record of Applicable Technical Directives
None

Figure 1. Example of NAVWPNSTA title page

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20. Section Screw Removal and Replacement. Missing, corroded, or damaged section screws shall be replaced as follows:

Support Equipment Required

Nomenclature	Part No./Type Designation	Cage
Multimeter	AN/USM-311	XXXXX
Hoist, engine/ wing removal	E289600 (E309707)	XXXXX

Materials Required

Nomenclature	Specification No./ Part No.
Compound, Antiseize High Temperture	MIL-A-907
Oil, Lubricating, Grade	MIL-L-6081

a. Rotate AUR to access damaged or corroded screw. If section screw is missing, rotate AUR to access hole and proceed to step d.

b. Obtain torque fixture (figure 4) to loosen corroded or damaged section screw as follows:

CAUTION

Torque bit must be in up position before attaching torque fixture to AUR; otherwise screw head damage may occur.

(1) Turn torque bit handle counterclockwise to raise torque bit.

CAUTION

Torque fixture adjustment shall be done by hand. The use of tools will damage the torque fixture.

(2) Turn clamp handle until loose.

(3) Open torque fixture and position around missile.

(4) Close torque fixture.

(5) Position bit over section screw.

(6) Secure torque fixture in place by lowering clamp handle and tightening.

CAUTION

Ensure bit is properly aligned with screw. Failure to do so may result in damage to screw head.

(7) Lower bit into screw head by rotating torque bit handle clockwise while holding bit in proper alignment with screw head.

CAUTION

When used for section screw removal, torque fixture is designed to loosen section screws. It is not designed to completely remove section screws. Use a speed wrench to remove the loosened screws.

(8) Snap 3/8 inch hinged handle socket wrench into torque fixture bit drive socket.

(9) Rotate wrench handle counterclockwise one or two turns.

(10) Raise bit by rotating torque handle counterclockwise until pins rest in "V" grooves.

(11) Turn clamp handle until loose. Unlatch torque fixture and remove.

c. Remove screw with speed wrench and torque bit.

CAUTION

Do not attempt to fully torque section screws with speed wrench. Use a speed wrench only for initial tightening. For final torquing, use the torque fixture.

d. Apply anti-seize compound to new section screw; insert in missile and tighten with speed wrench. If screw cannot be inserted, refer to nut or nut ring replacement (this WP). MAJOR

Figure 2. Example of support equipment required and materials required lists

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AIM-9H/L/M, ATM-9H-3, or NATM-9L/M AUR

Cookoff time	65 seconds
Withdrawal distance	1,800 feet

Warhead WDU-17/B, Mk 48 Mods

Cookoff time	150 seconds
Withdrawal distance	1,800 feet

Warhead WDU-9A/B, WDU-9/B

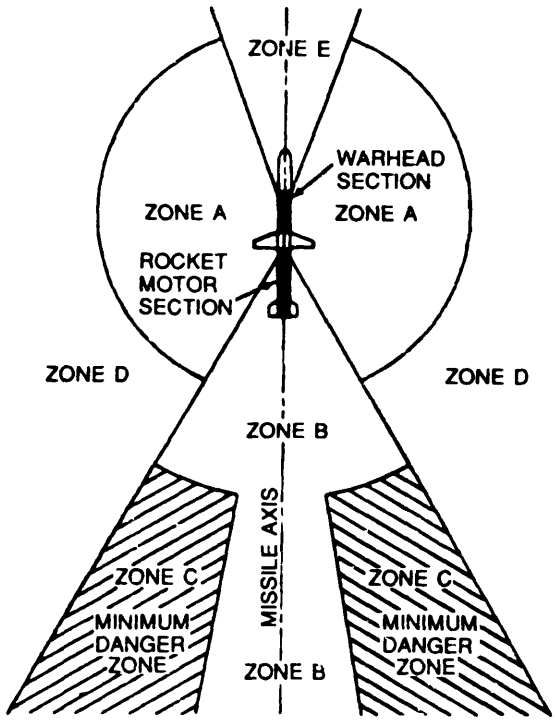
Cookoff time	100 seconds
Withdrawal distance	1,800 feet

Rocket Motor, Mk 36 Mods

Cookoff time	65 seconds
Withdrawal distance	150 feet

Figure 3. Example of cookoff time and evacuation distance

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ZONE	CHARACTERISTICS	RADIUS	ARC
A	LETHAL BLAST PRESSURE	11 FT MEASURED FROM CENTER OF WARHEAD SECTION	130° EACH SIDE OF MISSILE
B	LETHAL BLAST	(1) 15 FT MEASURED FROM FRONT OF ROCKET MOTOR SECTION	30° EACH SIDE OF MISSILE AXIS
		(2) 200 FT MEASURED FROM NOZZLE OF ROCKET MOTOR SECTION	10° EACH SIDE OF MISSILE AXIS
C	SAFEST APPROACH FOR FIREFIGHTING	NO CLOSER THAN 15 FT FROM FRONT OF ROCKET MOTOR SECTION	20° ± 10° (10° - 30°), MEASURED 1C° OFF EACH SIDE OF MISSILE LONGITUDINAL AXIS
D	LETHAL, FRAGMENTATION	2000 FT	130° EACH SIDE OF MISSILE
E	LETHAL, FORWARD THRUST OF MISSILE	NOT DETERMINED	20° EACH SIDE OF MISSILE AXIS

SHK004

Figure 1-4. Minimum Danger Approach Zones

Figure 3. Example of cookoff time and evacuation distance - continued

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Table 1. Missile Configuration Matrix

MISSILE CONFIGURATION TACTICAL	GCG						TDD						S A DEVICE				WARHEAD						
	MK 18 MOD 1 PN 2430934	MK 18 MOD 2 PN 2580665	MK 18 MOD 3 PN 2603940	MK 18 MOD 4 PN 2680000	MK 32 MOD 0 PN 1569426		MK 15 MOD 1 PN 2203479	MK 15 MOD 2 PN 1855947	MK 15 MOD 3 PN 2238168	MK 24 MOD 0 PN 2136149	MK 47 MOD 0 PN 2208359		MK 13 MOD 0 PN 1508436	MK 13 MOD 1 PN 1584368	MK 36 MOD 0 PN 2186050		MK 48 MOD 0 PN 1568432	MK 48 MOD 2 PN 1617384	MK 48 MOD 4 PN 2603878	WDU-8/B PN 2605168	MK 70 MOD 0 PN 2676137	MK 70 MOD 0 PN 1569434	
AIM 9D 1 PN 2605720-1	X										X	X			X	X			X	X	X		
AIM 9D 2 PN 2605720-2	X						X	X	X						X	X			X	X	X		
AIM 9G 1 PN 2605721-1		X									X	X			X	X			X	X	X		
AIM 9G 2 PN 2605721-2		X					X	X	X						X	X			X	X	X		
AIM 9H-2 PN 2605722-2			X				X	X	X						X	X			X	X	X		
AIM 9H-3 PN 2605722-3				X			X	X	X						X	X			X	X	X		
MISSILE CONFIGURATION TRAINING, EXERCISE																							
GDU-1/B PN 1562874					X							X				X						X	
ATM 9D 1 PN*	X											X										X	
ATM 9D 2 PN*	X										X	X			X	X					X		
ATM 9D 3 PN 2605059	X						X	X	X						X	X					X		
ATM 9D 5 PN 2605858-1	X						ALL COMPONENTS EXCEPT GCG ARE REPLACED BY MDU-26/A DUMMY BODY SECTION																
ATM 9G 1 PN*		X										X										X	
ATM 9G-2 PN*		X									X	X			X	X					X		
ATM 9G 3 PN 2606060		X					X	X	X						X	X					X		
ATM 9G-5 PN 2605858-2		X					ALL COMPONENTS EXCEPT GCG ARE REPLACED BY MDU-26/A DUMMY BODY SECTION																
ATM 9H 1 PN*			X									X										X	
ATM 9H 3 PN 2606061			X				X	X	X						X	X					X		
ATM 9H 9 PN 2605858-3			X				ALL COMPONENTS EXCEPT GCG ARE REPLACED BY MDU-26/A DUMMY BODY SECTION																

*TO BE ASSIGNED

Figure 4. Example of missile configuration matrix

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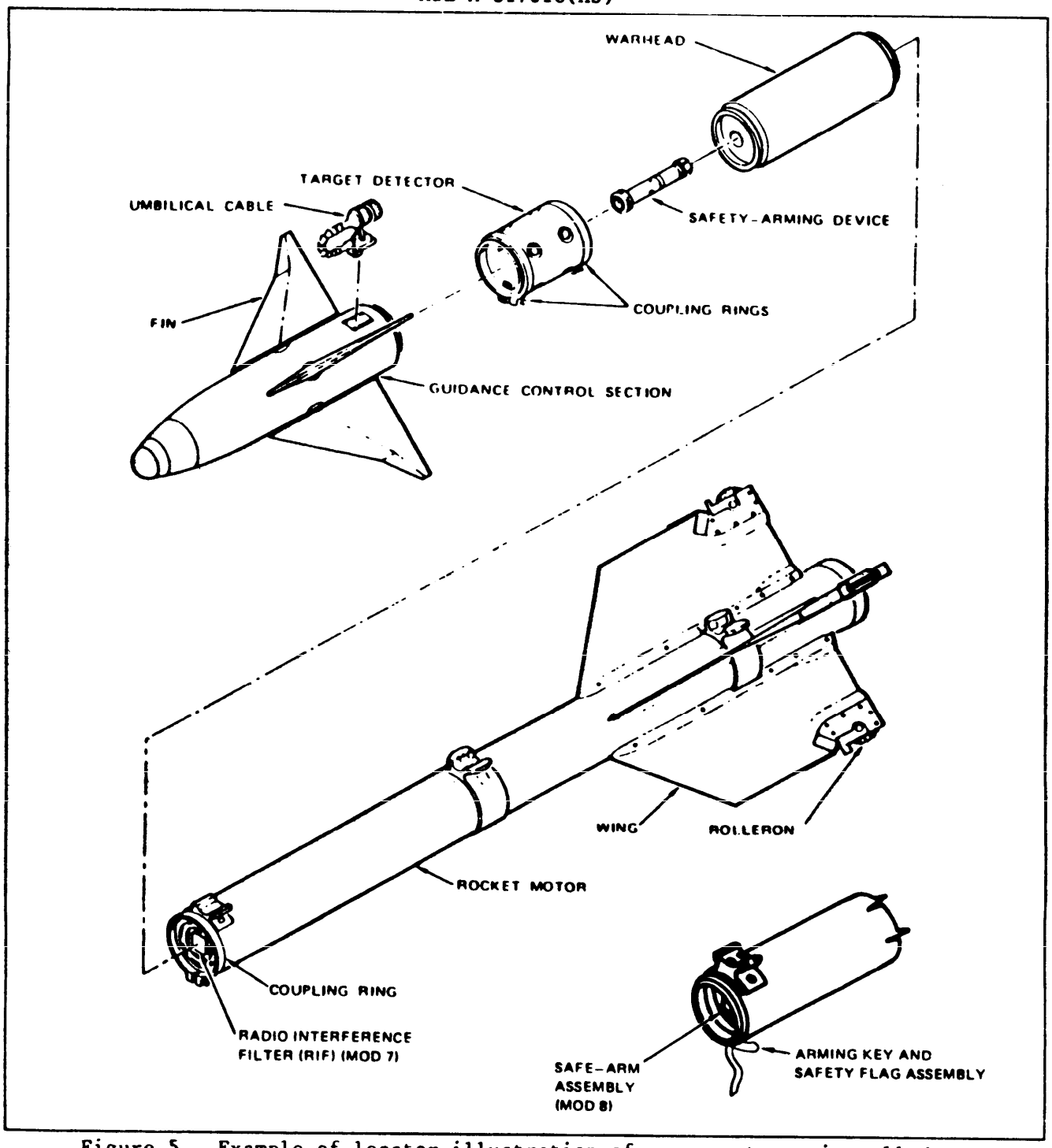
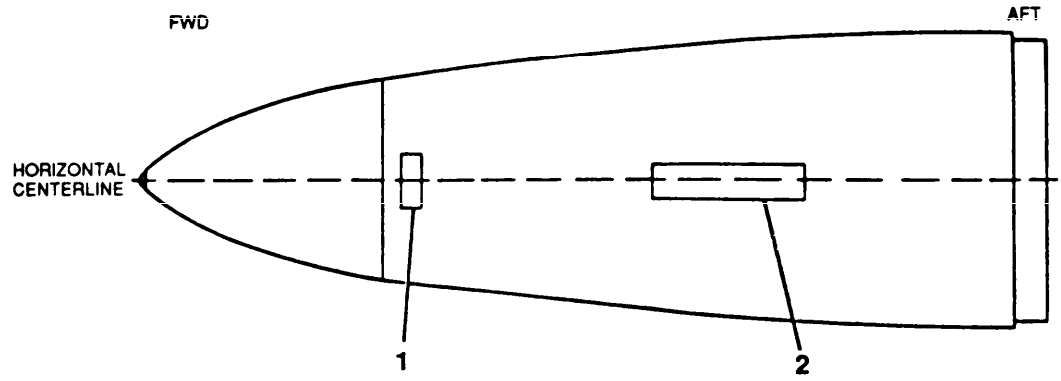
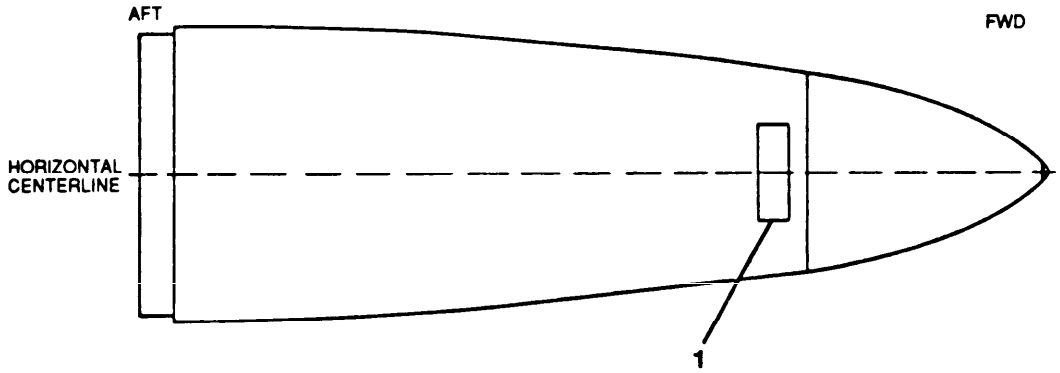


Figure 5. Example of locator illustration of components as installed in the end item

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(A) LEFT SIDE



(B) RIGHT SIDE

VIEW	ITEM	CONFIG	MARKING	CHARACTERISTICS
A,B	1	AGM-88A ATM-88A	NO LIFT FRAGILE RADOME 1.3"	1/8-INCH BLACK LETTERS ARROWS/BRACKET ARE 1/4-INCH BLACK LINE CENTERED PERPENDICULAR TO MISSILE HORIZONTAL CENTER LINE, 1 INCH AFT OF GUIDANCE/ RADOME INTERFACE
A	2	AGM-88A ATM-88A	NSN XXXX-XX-XXX-XXXX-XXXX GUIDANCE SECTION, WGU-2/8 SN PN XXXXX XXXXXXXXX	1/4-INCH BLACK LETTERS CENTERED ON MISSILE HORIZONTAL CENTER LINE

Figure 6. Example of inspection procedures

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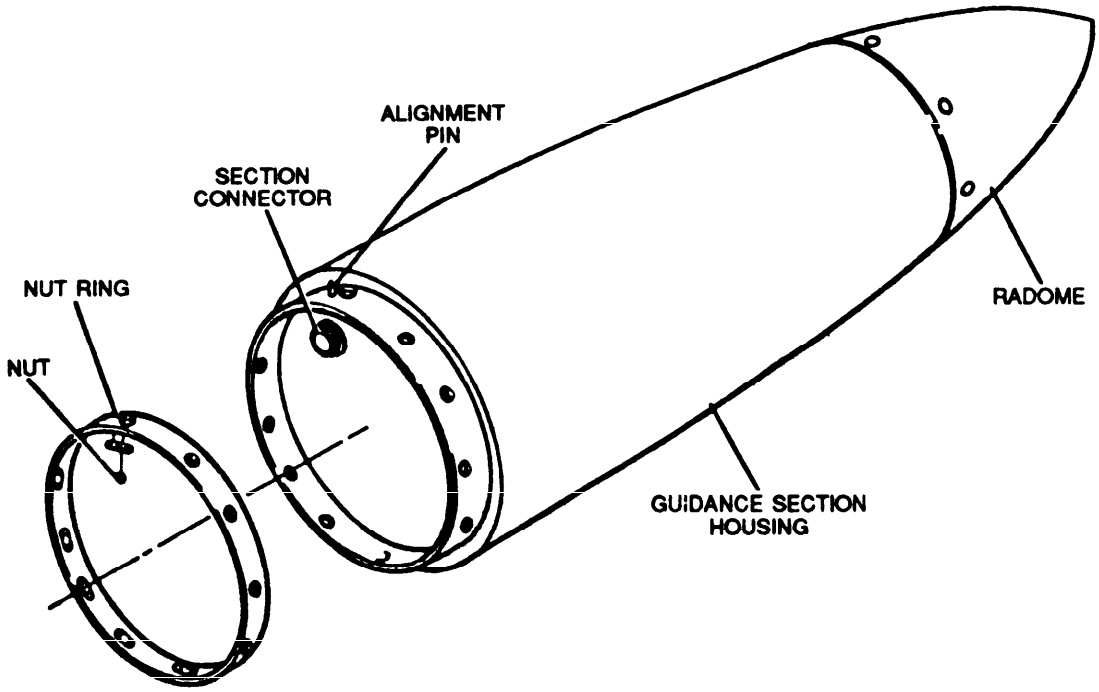


Figure 5. Guidance Section Inspection Points

Figure 6. Example of inspection procedures - continued

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Table 1. Guidance Section Inspection Criteria

STEP	INSPECTION POINT	INSPECTION CRITERIA	DISPOSITIONS	DEFECT CLASS
1	Markings	a. Missing, illegible, or incomplete.	a. Replace (WP 017 00).	MINOR
2	Radome	a. Dirt, grease, or scuffed.	a. Clean (WP 017 00).	MINOR
		b. Scratched, nicked, or pitted paint which does not disturb glass fibers.	b. Touch-up (WP 017 00).	MINOR
		c. Scratched, nicked, or pitted paint which does disturb glass fibers.	c. Through f. Reject guidance section (this WP).	MAJOR
		d. Gouged, cracked, punctured.		MAJOR
		e. Anything that visibly alters the radome fibers.		MAJOR
		f. Metal tip loose or missing.		MAJOR
3	Guidance section housing	a. Dirt or grease.	a. Clean (WP 017 00).	MINOR
		b. Scratched, missing paint, burred.	b. Touch-up (WP 017 00).	MINOR
		c. Pitted or corroded.	c. Repair (WP 017 00).	MAJOR
		d. Dent more than 0.060 inches deep.	d. Reject guidance section (this WP).	MAJOR
		e. Dent less than 0.060 inches deep.	e. Test with AUR (WP 012 00).	MAJOR
		f. Punctured, creased, or cracked.	f. Reject guidance section (this WP).	MAJOR
		g. Gouge less than .060 inches deep.	g. Test with AUR (WP 012 00) and repair gouge (WP 013 00).	MAJOR

Figure 6. Example of inspection procedures - continued

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(4) The ABORT key directs the test set to go to the end of the test sequence.

b. The operator is instructed by the VKT to perform three manual operations (NULL for azimuth, ROTATE, and NULL for elevation) during the test sequence.

c. Null the IR target simulator and rotating the GCS require the operator to move to the electronic control panel (ECP) and use the TARGET DRIVE RELEASE button, the CONTINUE button, and the three-digit display.

NOTE

The red STOP pushbutton removes electrical power only form the GCS. Turning off the circuit breaker on the test set removes all power from the test set.

d. The red STOP pushbutton is used in emergencies to stop testing and remove electrical power from the GCS during test sequence. Pressing the red STOP pushbutton is a reset function during self-test. To remove all power from test set, turn off test set circuit breaker.

14. GUIDANCE CONTROL SECTION TEST.

NOTE

The test set uses an automatic testing sequence requiring the operator to follow instructions displayed on the VKT screen. Before starting test, paragraph 13 shall be reviewed.

Table 3 lists tester error codes.

Table 4 lists AIM-9L test sequences.

Table 5 lists AIM-9M test sequences.

a. Verify the GCS is properly mounted and connected (paragraph 12).

b. Verify the servo gas bleed valve is completely closed (clockwise).

Inlet pressure must not exceed 3,500 psig.

c. Open Refrigerated Detector Unit (RDU) gas valve (fully counterclockwise). RDU gas pressure gauge shall indicate 2,400-2,750 psig.

d. Open servo gas valve (fully counterclockwise). Servo pressure gauge shall indicate approximately 1,200 psig.

e. If required, verify data collection magnetic tape is not in SAFE position (arrow points away from SAFE) and install cartridge in tape recorder.

f. If required, insert test set system software tape cartridge in VKT tape drive.

g. Press and release the BREAK key twice.

h. Press and release HOME PAGE key.

i. Press and release AUTO LOAD key. Wait one minute for tape to rewind.

j. VKT displays:

G = GCS TEST
S = SELF-TEST
C = CALIBRATION PROCEDURE
ENTER TYPE

k. Press and release G key.

l. Press and release RETURN key.

NOTE

There will be a slight delay while VKT searches for the correct program.

m. VKT displays:

MISSILE TYPE
H = AIM-9H
L = AIM-9L
M = AIM-9M
ENTER TYPE

n. Press and release appropriate key and then press and release RETURN key. VKT displays: PLEASE WAIT FOR TAPE DRIVE OPERATION.

o. If data tape if full, the operator must follow the VKT instructions. VKT displays:

DATA LOGGING TAPE IS FULL, PLEASE REPLACE WITH A NEW TAPE AND PRESS RETURN TO PROCEED.

When a blank new tape is inserted, the Auxiliary Tape Unit will read through the entire tape.

Figure 7. Example of testing and troubleshooting procedures

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Support Equipment Required

Nomenclature	Part No./Type Designation	Cage
Torque Bit	5/16 in.	XXXXX
Torque Wrench	0-100 in. oz.	XXXXX
Torque Wrench	0-50 in. lbs.	XXXXX

Materials Required

Nomenclature	Specification No. Part No.
Sealing Compound Grade C	MIL-S-22473
Blank tape cassette	704AS6316

1. INTRODUCTION.

2. This work package (WP) contains instructions for All-Up-Round (AUR) testing of CATM-88A. Figure 1 illustrates typical missile test flow.

3. GENERAL.

4. Besides performing self-test, the AN/DSM160 missile test set (MTS) is capable of performing missile diagnostic built-in-test (BIT -- BIT with fail codes), all function test, and missile cable (umbilical and battery jumper cables) test. The MTS can also perform missile BIT (pass/fail -- no fail codes), guidance section, and control section tests (sections must be integrated into an AUR), but this capability is not normally used at the naval weapon station (NWS) or mobile missile maintenance unit (MMMU). The test set is comprised of five units -- cart, hood, missile interface assembly (MIA), HP-85 remote terminal, and nine cables. The MTS is shown in figure 2. Missile cable test is performed on a replacement cable prior to installation on the AUR full-function testing. Full function testing is run for the following reasons:

a. AURs that have been captive flown as determined by the AUR elapsed time indicator (ETI) shall be full function tested. This is determined by comparing the ETI with the configuration summary sheet.

b. Date for AURs that are missing configuration summary sheets shall be obtained from Fleet Analysis Center (FLTAC).

c. AURs that have had sections separated or replaced shall be full function tested.

d. AURs which have less than nine months remaining before maintenance due date expires shall be full function tested.

e. AURs which have failed an inspection step requiring testing as disposition shall be full function tested.

5. AUR diagnostic BIT test is run on the AUR for the following reasons:

a. After passing full function test, the battery jumper cable shall be replaced and diagnostic BIT run to verify jumper continuity.

b. After removing, repairing or replacing the umbilical cable, diagnostic BIT shall be run.

c. After replacing the target detector (TDD), diagnostic BIT shall be run to check connector continuity.

6. Testing of the AUR shall be performed on the Assembly and Test Stand (ATS, figure 3). If AUR is not on ATS, refer to WP 007 00 or 008 00 for AUR uploading procedure.

7. SAFETY AND EMERGENCY PROCEDURES.

8. Safety and emergency procedures are contained in WP 003 00.

Figure 8. Example of pretest setup

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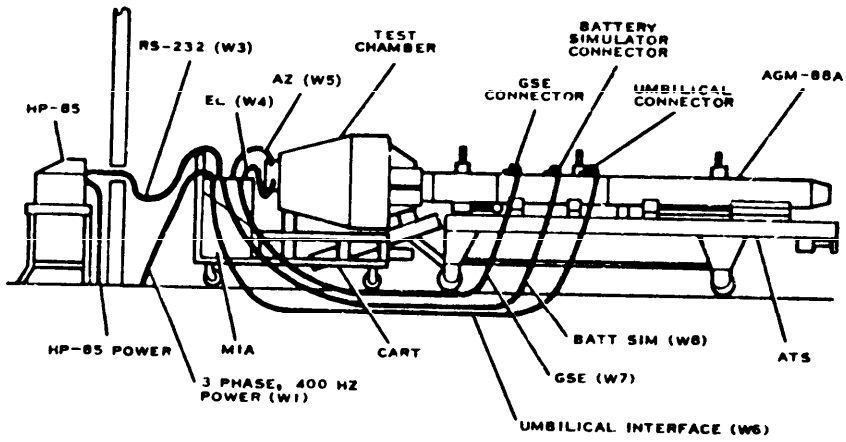


Figure 2. AUR Test Set

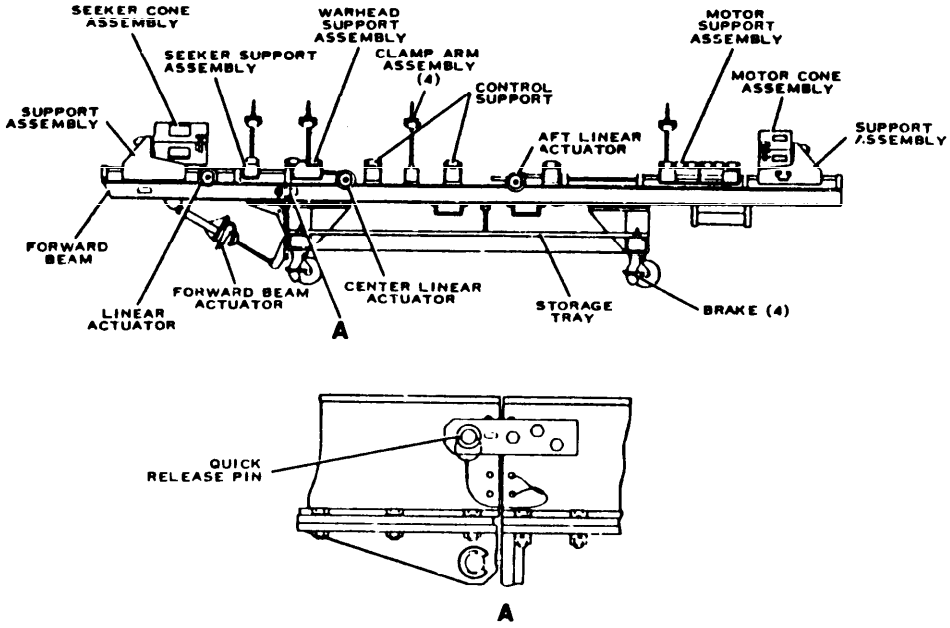
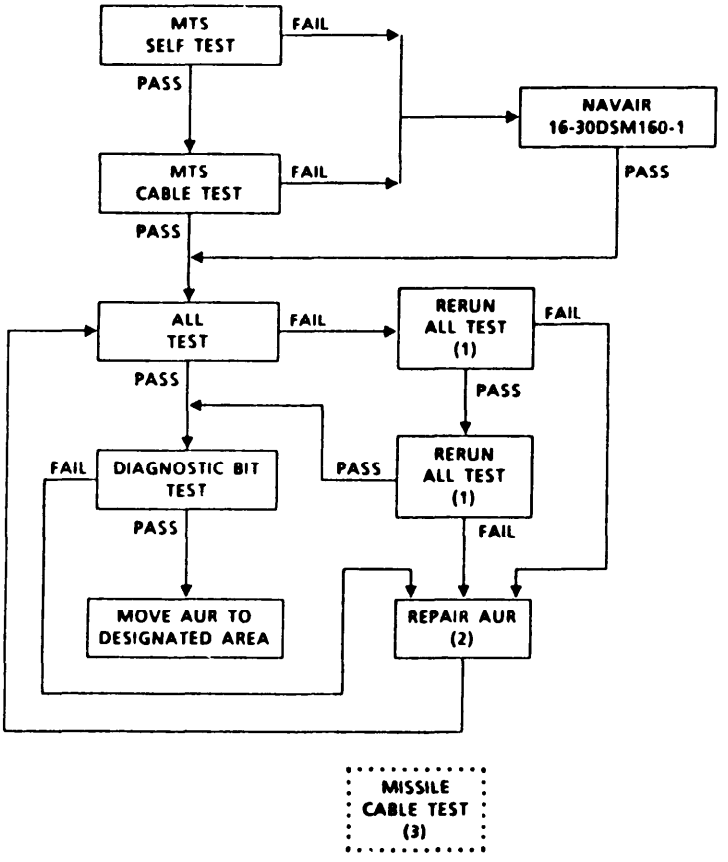


Figure 8. Example of pretest setup - continued

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(1) APPROPRIATE SECTION TEST MAY BE RUN IF DESIRED.

(2) SUBSEQUENT ALL TEST FAILURE OF SAME PARAMETER INDICATES HIGH PROBABILITY OF TEST SET FAILURE. RECOMMEND AN/DSM-160 SERVICEABILITY BE VERIFIED IN ACCORDANCE WITH NAVAIR 16-30DSM160-1. IF TEST SET IS DEFECTIVE, PREVIOUS REJECTS SHOULD BE RETESTED.

(3) MISSILE CABLE TEST IS REQUIRED ONLY WHEN SPECIFICALLY DIRECTED, OR WHEN A MISSILE CABLE IS REPLACED.

Figure 8. Example of pretest setup - continued

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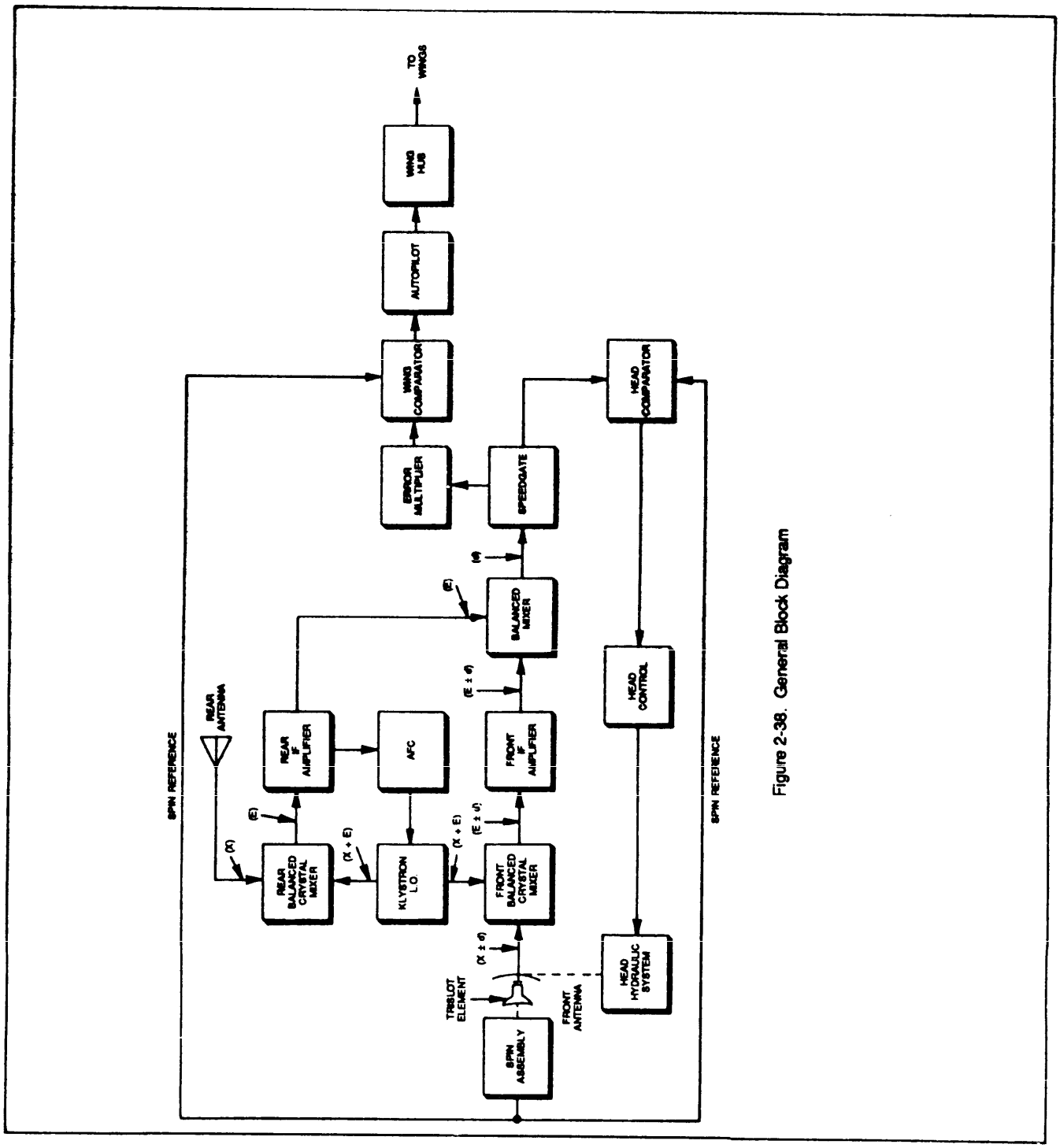


Figure 2-38. General Block Diagram

Figure 9. Functional flow diagram

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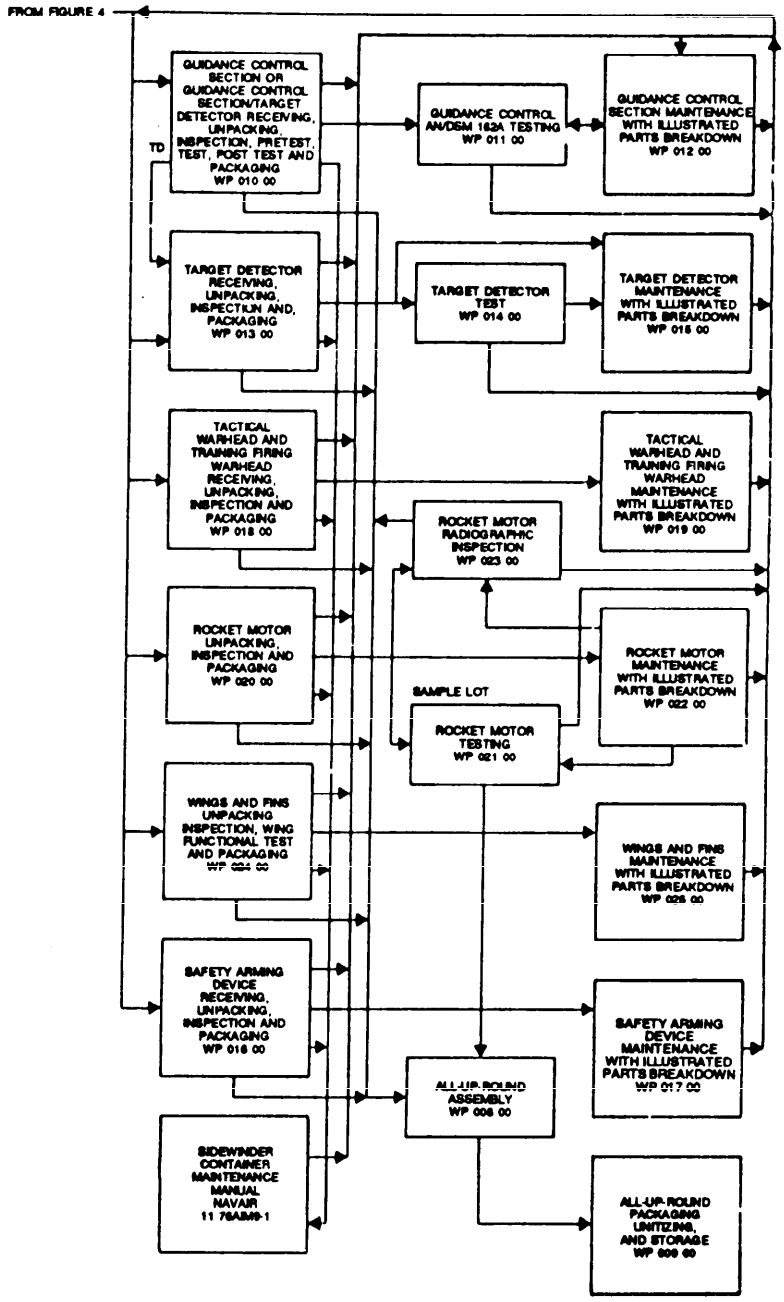


Figure 10. Example of processing flow diagram

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CAUTION

Damage to thermal coating could occur if wings are not placed in container in a vertical position. Ensure there is no binding.

b. Place wings vertically in container with lower and side cushion assemblies in place.

c. Ensure corrosion control has been performed on fins (paragraph 27).

d. Place plastic fin tray in container and ensure tray is fully seated. Place fins in fin tray.

e. Install top cushion on fins.

f. Lubricate sealing gasket with silicone compound.

g. Install cover and attach lead and wire seals to diagonally opposite locks. **(QA)**

h. Stencil markings (figure 7 or 8) on top and bottom of container with legible 1/4-inch high letters using black stencil ink or black lacquer. Cover with clear lacquer.

i. Complete and attach condition code tag in accordance with Appendix C of MIL-STD-129. Cover tag with clear lacquer. **(QA)**

16. PACKAGING, CONTAINER MK 418 MOD.

a. Obtain serviceable container Mk 418 Mod 0 (figure 9),

CAUTION

Damage to thermal coating could occur if wings are not placed in container in a vertical position. Ensure there is no binding.

b. Place wings vertically in container with lower and side cushion assemblies in place.

c. Lubricate sealing gasket with silicone compound.

d. Install cover and attach lead and wire seals to diagonally opposite locks. **(QA)**

e. Stencil markings (figure 9) on top and bottom of container with legible 1/4-inch high letters using black stencil ink or black lacquer. Cover with clear lacquer. **(QA)**

f. Complete and attach condition code tag in accordance with Appendix C of MIL-STD-129. Cover tag with clear lacquer. **(QA)**

17. PACKAGING, CONTAINER MK 430 MOD 0.

a. Obtain serviceable container Mk 430 Mod 0 (figure 10).

b. Place fins into molded plastic support tray inside container.

c. Lubricate sealing gasket with silicone compound.

d. Secure cover and stencil markings (figure 10) on top and one side of container with legible 1/4-inch high letters using black stencil ink or black lacquer. Cover with clear lacquer.

e. Attach lead and wire seal. **(QA)**

Figure 11. Example of quality assurance procedures - intermediate (fleet) maintenance manual

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